{EPITECH}

DAY 12_

< FILE DESCRIPTORS />



DAY 12

Preliminaries



Language: C
- All of your source files, except for useless files (binary, temp files, obj files,...), must be included in your delivery.
- Error messages have to be written on the error output, and the program should then exit with the 84 error code (0 if there is no error).



If you use your libmy for the following tasks, it must be built (if required) from the task's Makefile. The preferred way to do this is to call the lib's Makefile from the project's Makefile.



Allowed system function(s): `open`, `read`, `write`, `close`

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Task 01 - Cat

Delivery: cat/*

Write a program called `cat`, which executes the same tasks as your system's `cat` command. You do not have to handle options.
There is an unlimited number of files given as parameter.
`cat` without parameters must be supported.

You must deliver a `Makefile` with the following rules: `all`, `clean`, `fclean` and `re`, and it must not relink. The binary's name must be `cat`.
You may use the `errno` variable (refer to `man errno`), but `perror` and `malloc` are prohibited.
This task can only be performed by declaring a fixed-size array. It will have a limited size of approximately `30 kB`. In order to test the limitation, use the command limit in your shell:

```
Terminal
~$> limit stacksize 32
~$> limit stacksize
stacksize 32 kbytes
```



The read of size 1 is forbidden.



Limit is an internal feature of a specific shell. Find the good one :D.
`man cat`

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Task 02 - Testing cat

Delivery: tests/* cat/Makefile

You must now write unit tests for all the code composing your ”cat” program. We expect that most of your functions will be tested (so don't only test the final results).

Your tests will be built and executed with a rule `tests_run` that you have to add to your previous Makefile. But your tests files need to be in a directory called ”tests” **at the root** of your repository.
See Unit tests for C/C++ for more informations.



If you need to open a file (to obtain a file descriptor) for multiple tests, we encourage you to use fixtures to avoid code duplication.

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Task 03 - grep

Delivery: grep/*

Write a program called `grep`, which executes the same tasks as your system's `grep` command. You do not have to handle options.
There is an unlimited number of files given as parameter.

You must deliver a `Makefile` with the following rules: `all`, `clean`, `fclean` and `re`, and it must not relink. The binary's name must be `grep`.
You may use the `errno` variable (refer to `man errno`), but the `perror` function is prohibited.



For this task, and this task only, `malloc` and `free` are allowed.



You don't have to handle regex. We are only asking for a simple matching system.
`man grep`



The read of size 1 is forbidden.

Terminal

```
~$> ./grep looneytunes /etc/passwd
looneytunes:x:1000:100:looney tunes:/home/looneytunes:/bin/bash
~$> /grep http /etc/services
http 80/tcp
...
~$> /grep "application/pdf" /usr/share/misc/magic
!:mine application/pdf
~$> /grep http /doesnt_exist > /dev/null
grep: /doesnt_exist: No such file or directory
~$> /grep http /root > /dev/null
grep: /root: Permission denied
```


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Task 04 - Testing grep

Delivery: tests/* grep/Makefile

You must now write unit tests for all the code composing your `grep` program. We expect that most of your functions will be tested (so don't only test the final results).

Your tests will be built and executed with a rule `tests_run` that you have to add to your previous Makefile. But your tests files need to be in a directory called "tests" **at the root** of your repository.
See Unit tests for C/C++ for more informations.



If you need to open a file (to obtain a file descriptor) for multiple tests, we encourage you to use fixtures to avoid code duplication.



v 3

